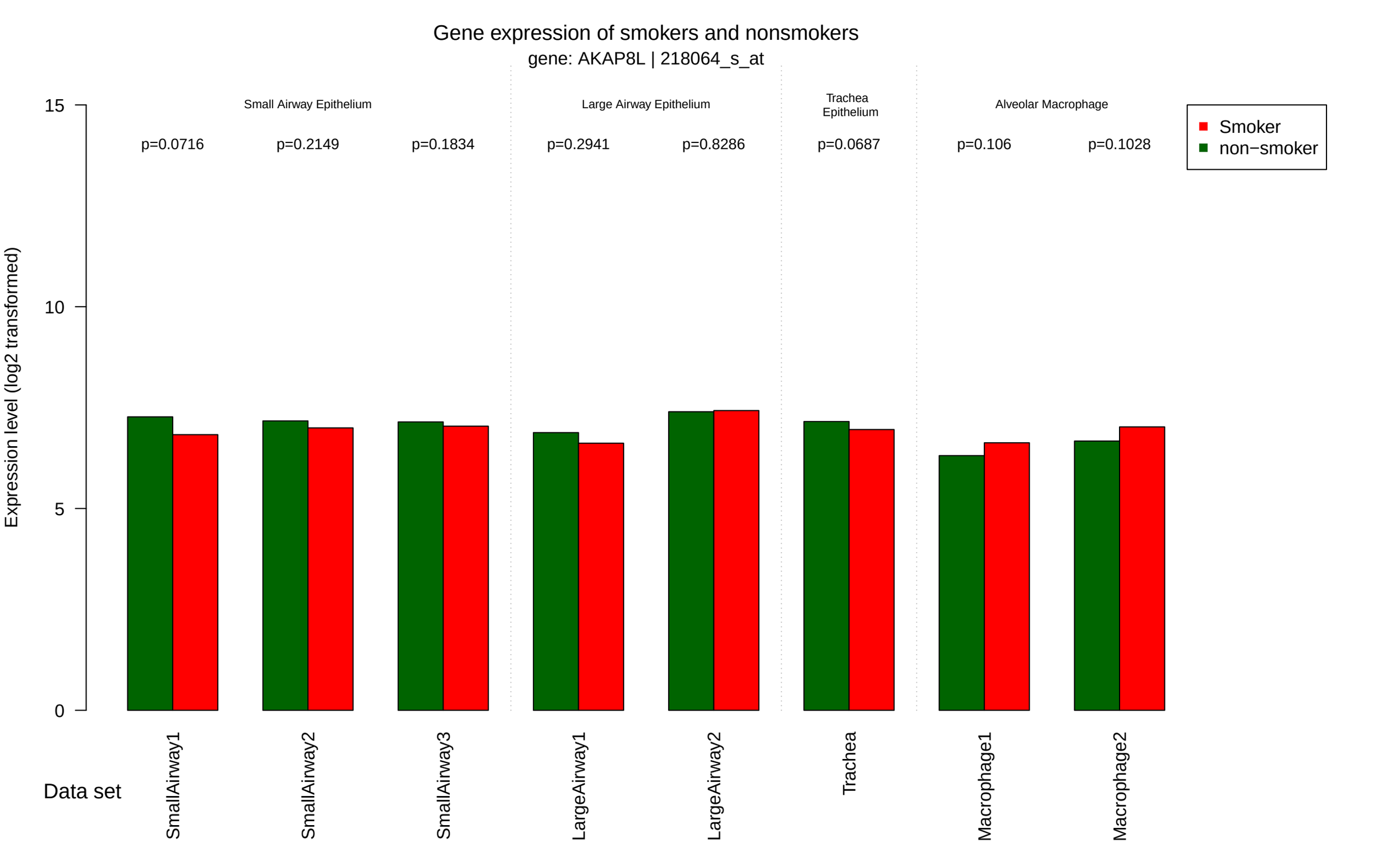
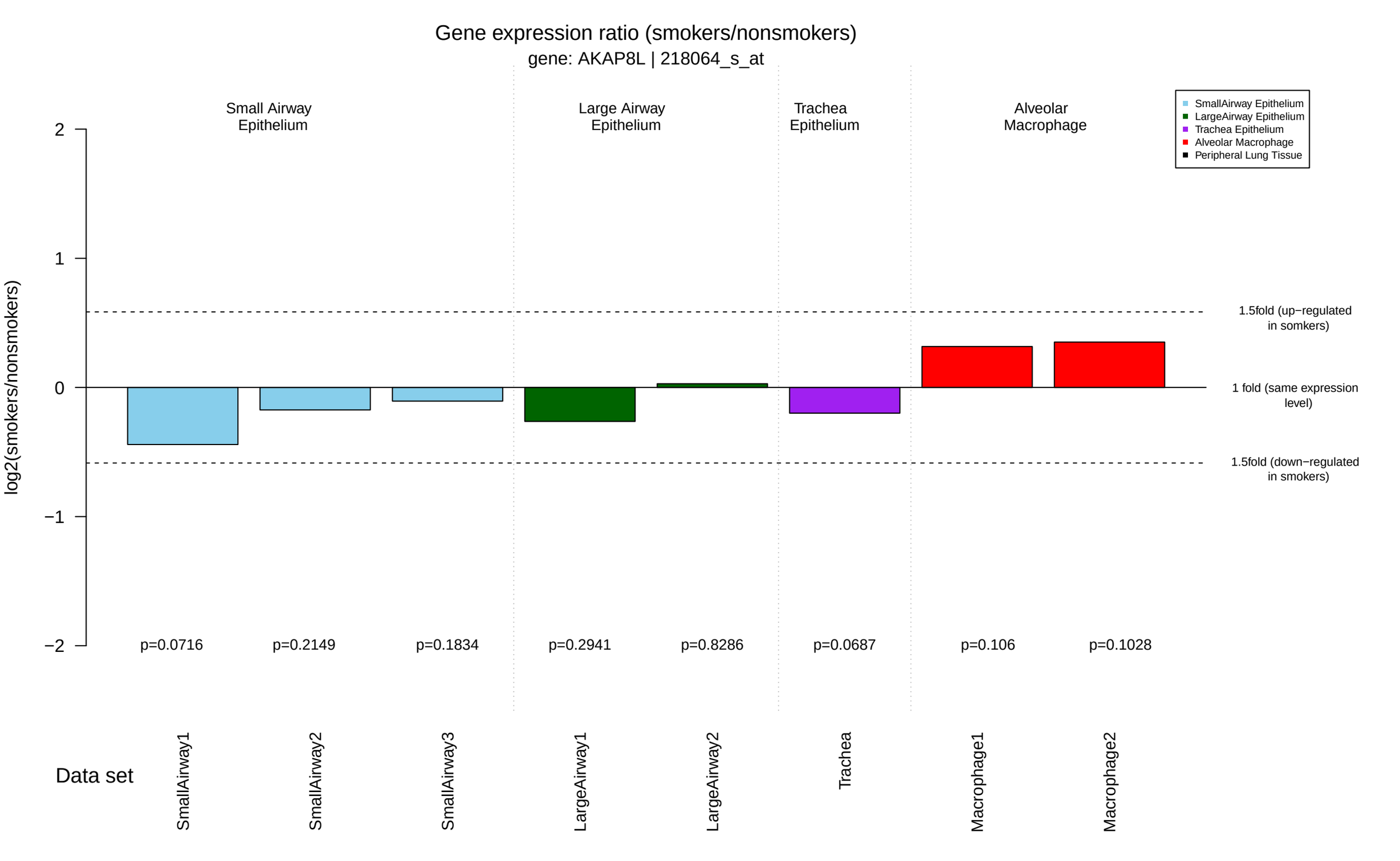
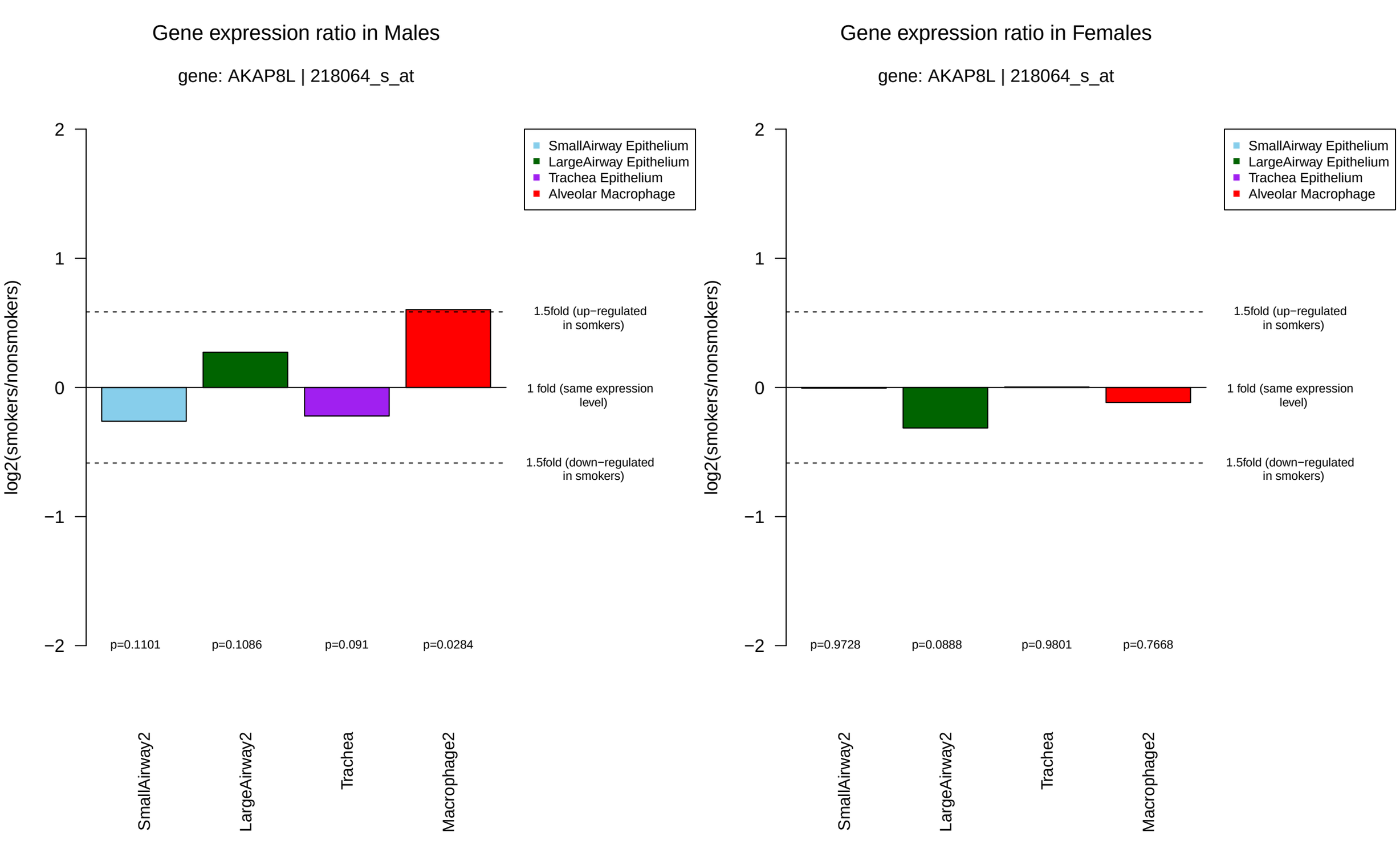
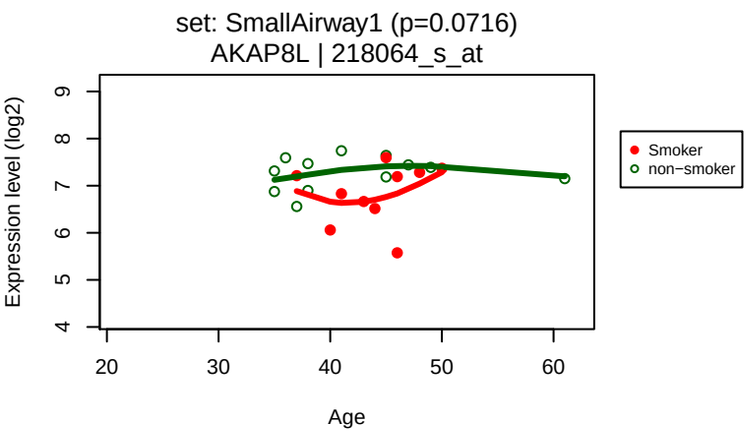
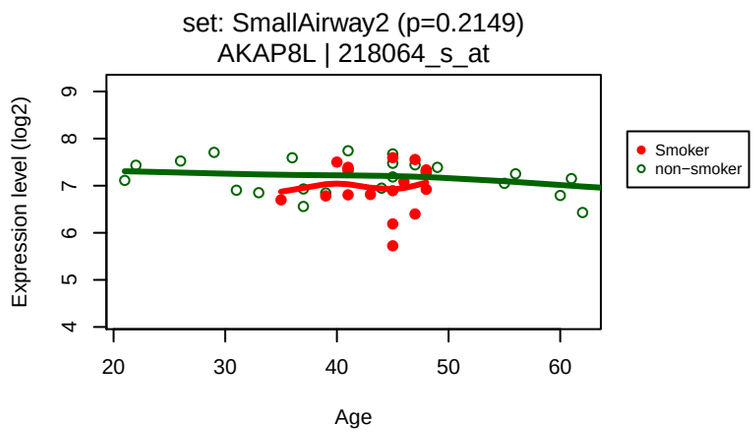
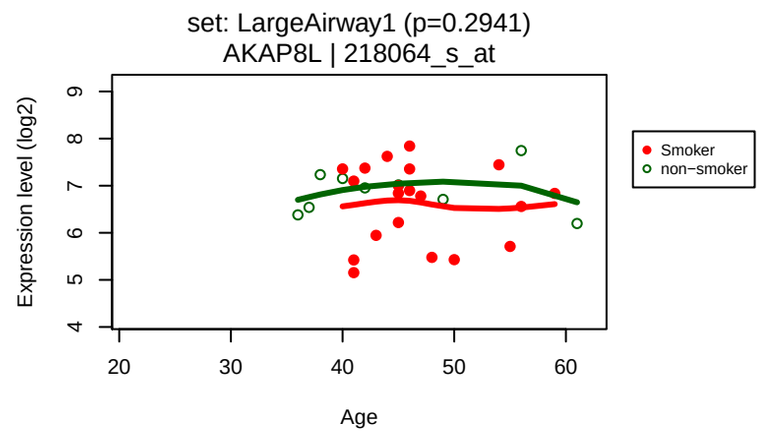

Gene expression of smokers and nonsmokers
gene: AKAP8L | 218064_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0716
p=0.2149
p=0.1834
p=0.2941
p=0.8286
p=0.0687
p=0.106
p=0.1028
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

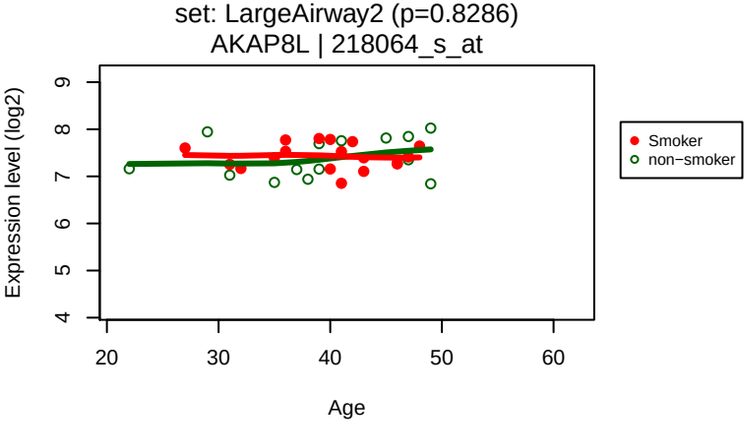

Gene expression ratio (smokers/nonsmokers)
gene: AKAP8L | 218064_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0716
p=0.2149
p=0.1834
p=0.2941
p=0.8286
p=0.0687
p=0.106
p=0.1028
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

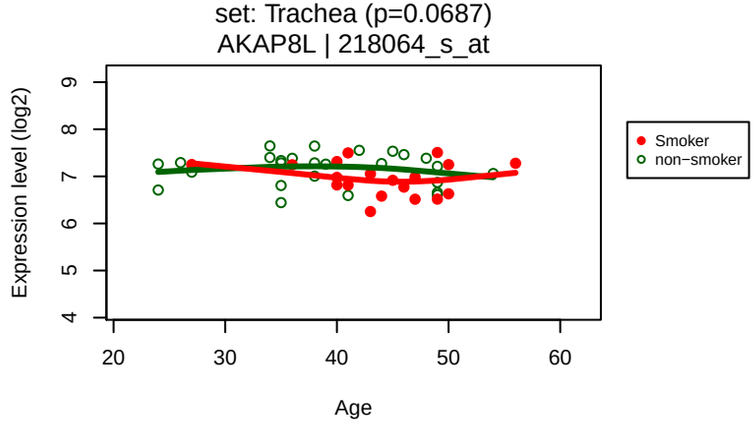

Gene expression ratio in Males
gene: AKAP8L | 218064_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1101
p=0.1086
p=0.091
p=0.0284
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP8L | 218064_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9728
p=0.0888
p=0.9801
p=0.7668
SmallAirway2
LargeAirway2
Trachea
Macrophage2

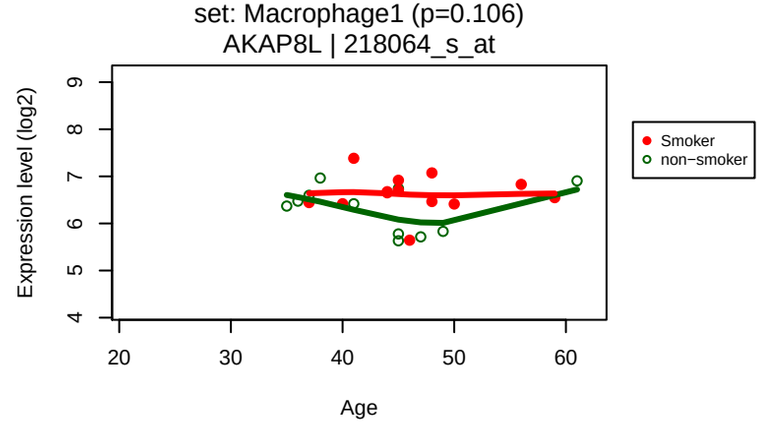

set: SmallAirway1 (p=0.0716)
AKAP8L | 218064_s_at
Expression level (log2)
Age
Smoker
non−smoker

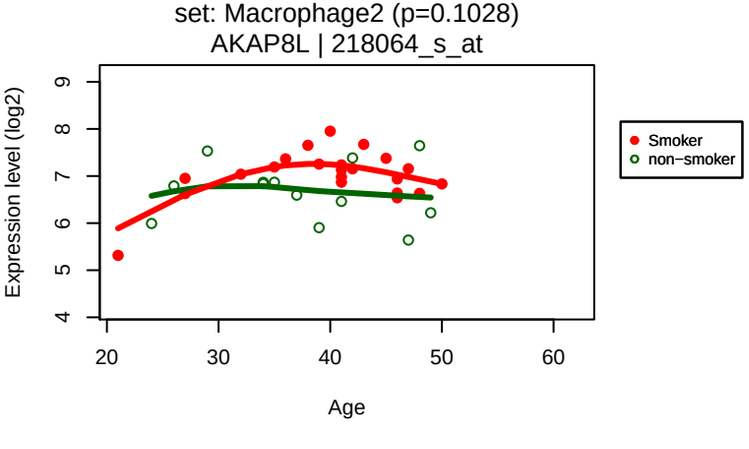

set: SmallAirway2 (p=0.2149)
AKAP8L | 218064_s_at
Expression level (log2)
Age
Smoker
non−smoker

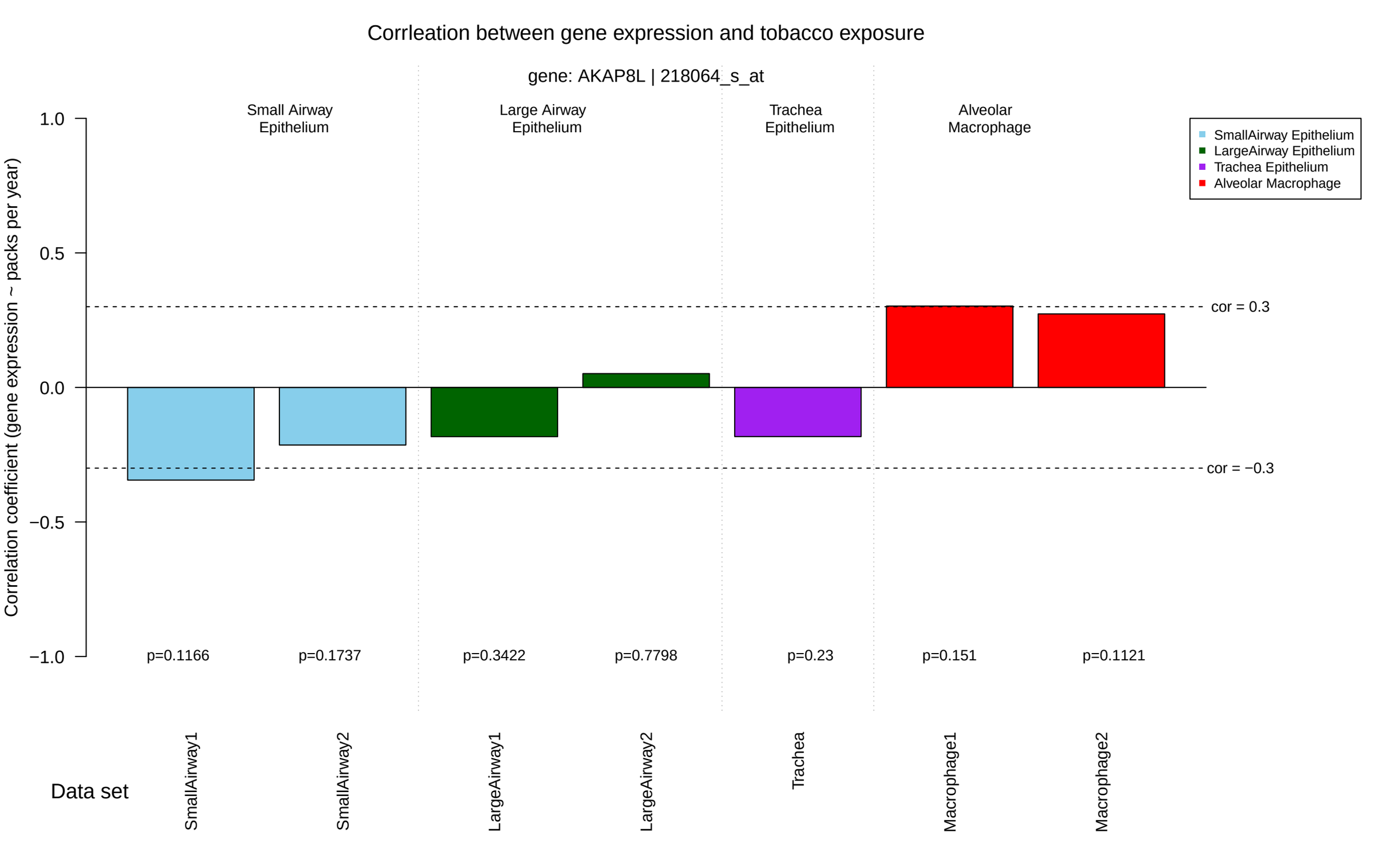

set: LargeAirway1 (p=0.2941)
AKAP8L | 218064_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8286)
AKAP8L | 218064_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0687)
AKAP8L | 218064_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.106)
AKAP8L | 218064_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1028)
AKAP8L | 218064_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP8L | 218064_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1166
p=0.1737
p=0.3422
p=0.7798
p=0.23
p=0.151
p=0.1121
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2